

NCT06023589 Efficacy Prediction Analysis

For NCT06023589, a principled efficacy-prediction ("PTRS") assessment lands in the "high probability of a positive primary efficacy result" bucket, given the drug's adult data, mechanism, and the pediatric Phase 3 design.[cite:38][cite:41][cite:45]

Below is the appraisal as if this were an internal research agent run, followed by an explanation of each step and the rationale.

Step 1 – Define the question and label

Target question:

$P(\text{trial meets its primary efficacy endpoint})$ for NCT06023589: HORIZON, pediatric tezepelumab in severe uncontrolled asthma.[cite:38][cite:45]

This trial is ongoing with an estimated primary completion in 2028, so this is an ex-ante prediction; there are no actual HORIZON results yet.[cite:45]

Step 2 – Extract structured trial features

From AstraZeneca and registry mirrors:[cite:38][cite:40][cite:45][cite:48]

- Phase: Phase 3 interventional treatment trial.[cite:38][cite:45]
- Indication: Severe uncontrolled asthma in children 5 to <12.[cite:40][cite:46]
- Study design:
 - Multicentre, randomized, double-blind, parallel-group, placebo-controlled.[cite:38][cite:45][cite:51]
 - 4–6 week screening/run-in, 52-week double-blind treatment, 12-week follow-up.[cite:38][cite:45][cite:51]
- Intervention:
 - Tezepelumab (human IgG2 λ monoclonal antibody against TSLP), subcutaneous.[cite:40][cite:45]
 - Comparator: placebo, subcutaneous.[cite:40]
- Population:
 - Age: 5 to <12 years.[cite:40][cite:45]
 - Severe uncontrolled asthma on medium- to high-dose ICS plus additional controller(s).[cite:40]
 - Key exclusions: other serious lung disease (e.g., cystic fibrosis, primary ciliary dyskinesia), major comorbidity, recent severe exacerbation requiring systemic steroids, recent ICS dose change, life-threatening asthma history requiring intubation/ventilation.[cite:40]

- Sample size: Planned N $\approx$ 231 participants.[cite:45]

These become the core “trial design” and “population” features for prediction.

Step 3 – Build a prior for tezepelumab efficacy

All available efficacy evidence for tezepelumab in severe asthma is pulled in and translated into a prior on the treatment effect.

Key sources:

- PATHWAY (Phase 2b) and NAVIGATOR (Phase 3) in adults/adolescents with severe uncontrolled asthma:
 - Overall annualized asthma exacerbation rate (AAER) reduction vs placebo $\approx$ 60% (rate ratio 0.40, 95% CI 0.34–0.48).[cite:41]
 - NAVIGATOR Japanese subset: AAER 1.54 vs 3.12 (rate ratio 0.49, 51% reduction).[cite:52]
 - Benefits across a wide range of eosinophil counts, FeNO levels, and allergen sensitivities (T2-high and T2-low), so effect is not narrowly biomarker-dependent. [cite:41][cite:57]
- Long-term DESTINATION extension:
 - Sustained reductions in exacerbations over 104 weeks with rate ratios around 0.42–0.61 vs placebo and stable safety.[cite:43][cite:47]
- Real-world PASSAGE study:
 - AAER decreased by 76% from baseline in diverse US patients with severe uncontrolled asthma treated with tezepelumab, with consistent reductions across phenotypes and underserved subgroups.[cite:53]
- Meta-analysis:
 - Pooled analysis shows significant AAER reduction (OR 0.67), meaningful ACQ-6 and FEV1 improvements, and similar or lower adverse event rates vs placebo.[cite:56] [cite:55]

From this, a research agent would set a prior on the true effect in severe asthma roughly as:

- Adult/adolescent AAER rate ratio $\theta_{\text{adult}} \approx 0.40\text{--}0.50$. [cite:41][cite:52]
- Very high confidence that $\theta_{\text{adult}} < 0.75$ (i.e., at least $\sim$ 25% reduction). [cite:41][cite:56]

For pediatrics, the effect is down-weighted to reflect ontogeny and phenotype uncertainty, giving a pediatric prior like:

- Pediatric prior mean rate ratio $\theta_{\text{ped}} \approx 0.50\text{--}0.60$ (40–50% reduction), with a 95% credible interval maybe spanning 0.35–0.80.

The exact numbers are modeling choices, but conceptually, this prior is: “strong, sustained, mechanism-consistent effect in adults; plausibly somewhat attenuated but still large in children.”

Step 4 – Combine prior with pediatric design to get power

The next step is to look at whether NCT06023589's design can detect that effect.

By analogy with prior tezepelumab pivotal trials and standard severe asthma designs:[cite:41][cite:43][cite:44][cite:55]

- Likely primary endpoint: annualized asthma exacerbation rate over the 52-week treatment period.[cite:41][cite:43][cite:44][cite:51]
- Baseline AAER in severe uncontrolled pediatric populations is typically >1/year on high-dose ICS and controllers; prior biologic pediatric trials often assume ~1.5–3 events/year in placebo arms. Direct numbers for this protocol are not published, but the HORIZON design suggests frequent events are assumed.[cite:41][cite:55]
- Statistical model: negative binomial regression for AAER, as in PATHWAY/NAVIGATOR.[cite:41][cite:52][cite:55]
- Sample size: $N \approx 231$ randomized children, presumably about 1:1 tezepelumab:placebo.[cite:45]

Given that:

- If true pediatric rate ratio $\theta_{\text{ped}} \approx 0.55$, an N of about 230 with 1 year of follow-up and high baseline event rates is typically adequate to achieve at least 80–90% power at two-sided alpha 0.05 for AAER. This mirrors what was done in adult tezepelumab trials, which were designed for similar or slightly smaller rate ratios at somewhat larger N.[cite:41][cite:52][cite:55]
- The multicentre, randomized, double-blind, placebo-controlled design reduces bias and variance, improving the precision of the estimated rate ratio.[cite:38][cite:45][cite:51]

So, conditional on the prior effect size, the conditional probability that the trial detects that effect, meaning achieves statistical significance on the primary endpoint, is high.

Step 5 – Adjust for pediatric-specific uncertainties

A research agent would then apply pediatric-specific discount factors.

Possible downward pressures on efficacy signal:

- Biological/phenotypic heterogeneity:
 - Pediatric asthma includes phenotypes like viral-induced wheeze that may be less TSLP-driven than classic eosinophilic/allergic severe asthma.[cite:55]
 - This could shrink the average rate ratio toward 0.6–0.7 if a sizable fraction of enrolled children are in phenotypes less responsive to TSLP blockade.
- Operational variability:
 - Adherence to inhaled therapy and visit schedules is more challenging in pediatrics and may add extra noise to exacerbation endpoints, biasing effect estimates toward the null.[cite:6][cite:29]

Mitigating factors:

- HORIZON inclusion criteria, severe uncontrolled on high-dose ICS plus controller, and exclusion of recent medication changes and life-threatening events select a high-risk, more homogeneous severe population.[cite:40]
- Adult data show efficacy across biomarker and phenotype strata, suggesting TSLP is upstream enough to capture multiple pediatric phenotype clusters.[cite:41][cite:57]

Net: the agent might knock the prior effect mean from, say, a 45–50% reduction down to a conservative 35–40% (rate ratio about 0.60–0.65) to reflect pediatric uncertainty, but not all the way to near-null.

Step 6 – Translate to a quantitative predicted success probability

Using published global PoS studies and AI-based predictors as calibration:[cite:14][cite:26][cite:27][cite:28][cite:33][cite:35]

- Historical Phase 3 respiratory/immune trials have technical success rates, approval or primary endpoint success, around 50–60%, depending on mechanism and sponsor.[cite:14]
- Trials with the same asset already approved in closely related patient populations and indications, a mechanism with strong, replicated Phase 2/3 data and favorable safety, and a straightforward pediatric label-extension-like design generally sit at the top end of PoS distributions in ML/AI models such as inClinico and MIT’s machine-learning PoS models. [cite:27][cite:28][cite:33]

Given NCT06023589’s features:[cite:38][cite:40][cite:45][cite:46][cite:51]

- Strong positive factors:
 - Same drug already proven and approved in adults/adolescents with severe uncontrolled asthma.[cite:41][cite:42][cite:43]
 - Mechanism upstream in type 2 inflammation, effective across biomarker-defined phenotypes.[cite:41][cite:57]
 - Phase 3, randomized, double-blind, placebo-controlled, 1-year treatment; good internal validity.[cite:38][cite:45]
 - Adequate N for expected effect size and endpoint.[cite:45][cite:41][cite:55]
- Mild negative or uncertain factors:
 - Pediatric population with potentially more heterogeneity in asthma pathophysiology. [cite:55]
 - Operational noise in pediatric studies.[cite:6][cite:29]

Putting these into a notional PoS/PTRS output, a reasonable research-agent prediction, technical efficacy success and not regulatory, would be:

- Predicted probability that NCT06023589 meets its primary efficacy endpoint: on the order of about 0.75–0.90, meaning confidently above average Phase 3.

This can be thought of as baseline Phase 3 respiratory PoS around 0.55, multiplied by a strong positive adjustment for “same mechanism already successful in near-identical indication,”

slightly reduced for pediatric uncertainties, netting high 70s to 80s.[cite:14][cite:27][cite:28][cite:33]

Because the exact analysis dataset or the proprietary model weights are not available, this report gives a calibrated range rather than a point estimate. Methodologically, however, this is the same thing those agents do: structured feature extraction plus mechanism-specific prior plus historical calibration.[cite:26][cite:27][cite:28][cite:33][cite:35]

Step 7 – Verbal conclusion for NCT06023589

Putting all of the above into a concise statement:

- Given the strong adult/adolescent tezepelumab efficacy signal across phenotypes, sustained long-term benefit, and replicated real-world data, the biological prior for efficacy in severe pediatric asthma is high, though modestly discounted for developmental and phenotype differences.[cite:41][cite:43][cite:53][cite:55][cite:57]
- The HORIZON Phase 3 design, NCT06023589, is rigorous, well-powered, and closely mirrors successful adult programs, increasing the chance that any true effect will be detected. [cite:38][cite:41][cite:45][cite:52][cite:55]
- A research-grade PTRS model fed these features would assign a high probability that HORIZON will show a statistically significant reduction in exacerbations versus placebo at 52 weeks, likely in the upper quartile of Phase 3 PoS estimates.[cite:14][cite:26][cite:27][cite:28][cite:33][cite:35]

For portfolio-style appraisal, NCT06023589 scores as meaningfully above a typical Phase 3 respiratory program and can be treated as a relatively low-risk technical efficacy bet versus the broader clinical trial universe.[cite:14][cite:41][cite:45]